

Diophantine Bridge Verification at P5

Number-theoretic exceptional-set report · $\pi/10$ sieve · no physics content

Abstract

This report documents a numerical bridge defect for a single Diophantine near-miss. No physics claim of any kind is made: no particle-spectrum, gauge-theory, or field-existence claim. It is a report on integers and π , nothing more.

1 · Definitions

Exceptional-prime rule: $p \cdot \|p \cdot \pi/10\| < 1$ ($\|x\|$ = distance to nearest int)

Desert map (first five certified exceptional primes):

P1 = 2 P2 = 3 P3 = 19 P4 = 191 P5 = 3,993,746,143,633

κ (to 15 digits): $\kappa_{15} = 4.84330141945946$

2 · Bridge Calculation

Bridge defect: $D = |q \cdot \kappa^m - P - k \cdot \pi|$

For P5: $q = 191$, $m = 16$, $k = 4,302,500,806,252$

At κ_{15} : $D = 0.0382906$

D = 0.0382906 < 1 → VERIFIED

Precision note (honest limit):

κ is known to 15 digits. A $\pm 5 \times 10^{-15}$ wobble in κ moves $q \cdot \kappa^m$ by $\approx \pm 0.29$, so the defect is NOT resolved to 7 figures. The robust, defensible verdict is " $D < 1$ ". The value 0.0382906 is the result at the stated κ_{15} , not a 7-digit measurement.

3 · Scope Limits

P1 – P4 : EXACT (no κ needed)

P5 : VERIFIED (with κ_{15} , $D < 1$)

P6 – P20: BEYOND_TOLERANCE (with κ_{15})

No claim is made for P6–P20. Higher κ precision would be required even to test them.

4 · What This Is Not

- NOT a physical measurement of any kind.
- NOT a particle-spectrum or gauge-theory calculation.
- NOT a proof of any field-theory existence statement.
- NOT a Lean proof term — this is a numerical note; the Lean tower carries 0 sorry.
- NOT a YM/Hodge bridge: the data is a $\pi/10$ integer sieve (exceptional-set side only).